

Uncertainty quantification in LLMs

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What is uncertainty?

Life is uncertain



Uncertainty in daily life examples

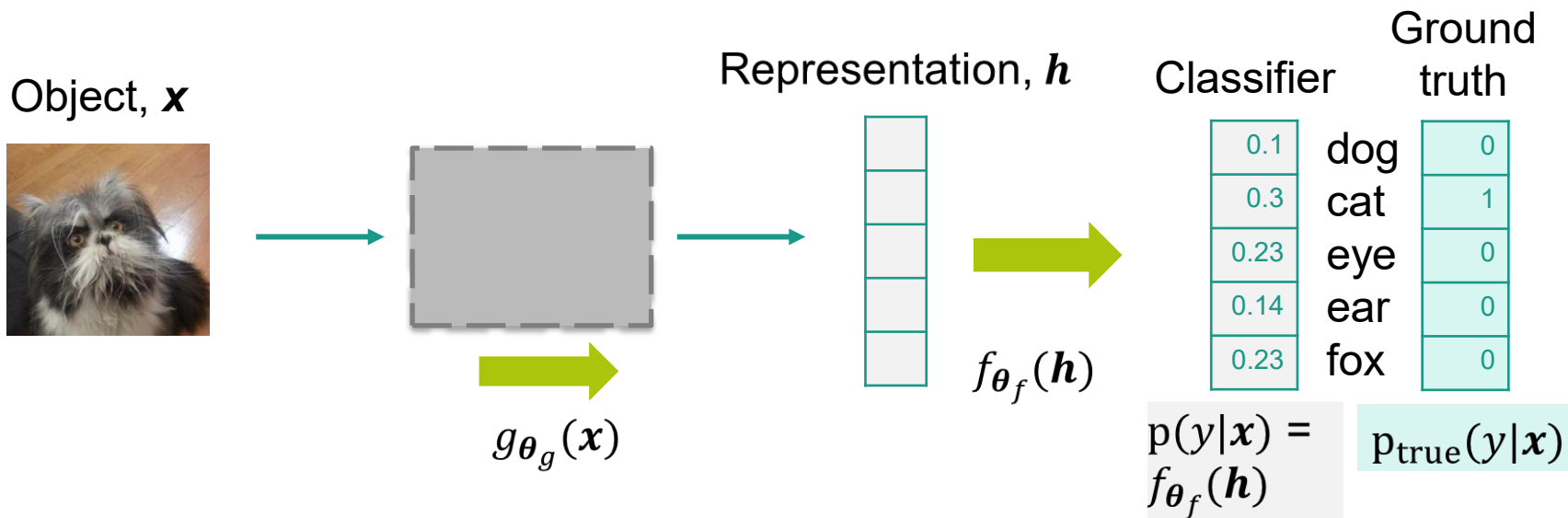
- **Weather forecasting** - is it going to rain?
- **Investment decisions** - will the stock go up or down?
- **Medical diagnosis** - what's the exact disease?

What is Uncertainty?

- **Uncertainty** refers to situations involving imperfect or unknown information.
- It's the gap between what we know and what we don't know.

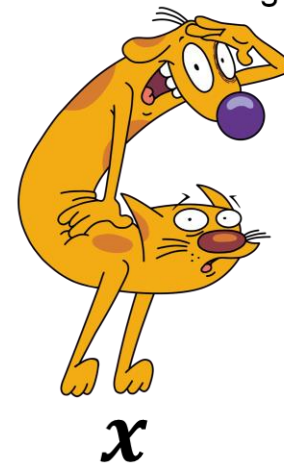
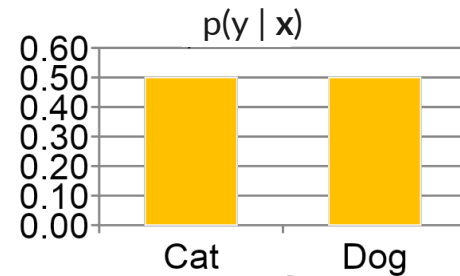
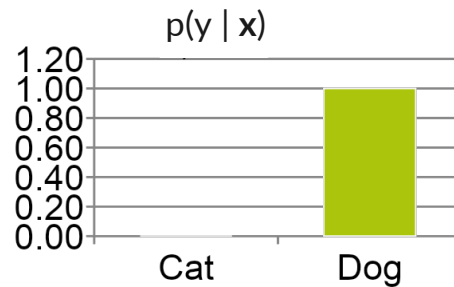
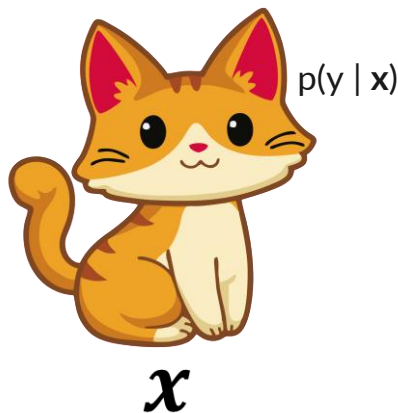
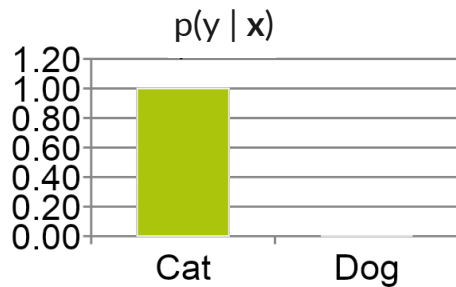
Uncertainty estimation

We already have probabilities in the model



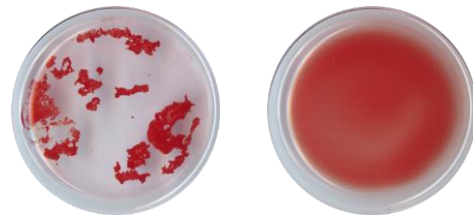
We maximize the similarity of output probabilities with a one-hot encoding

We (almost) never know the uncertainties in the ground truth



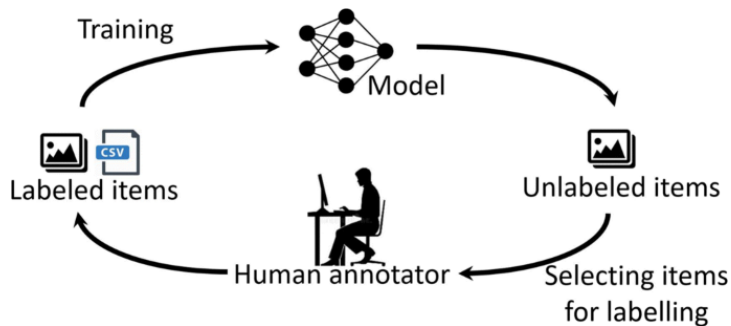
Uncertainty in applications

Medical diagnosis – should we retake a performed blood test?

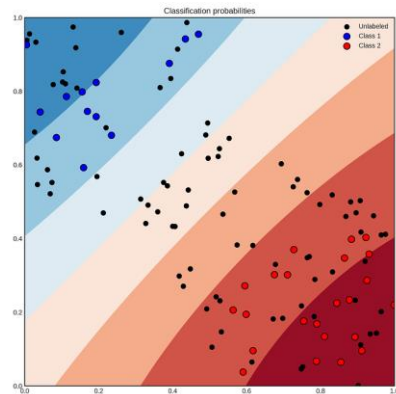


Source: Synteco, IITP RAS

Active learning – we should label data with high uncertainty from a pool of unlabeled samples

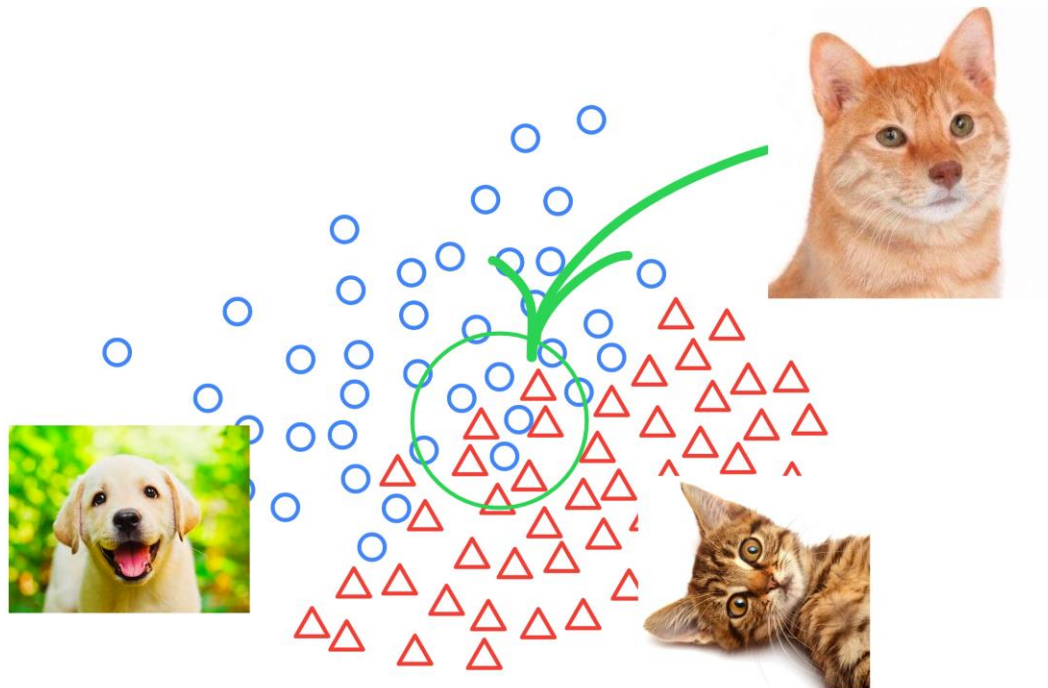


Source: <https://twitter.com/1evilidiot/status/794613309613821952>



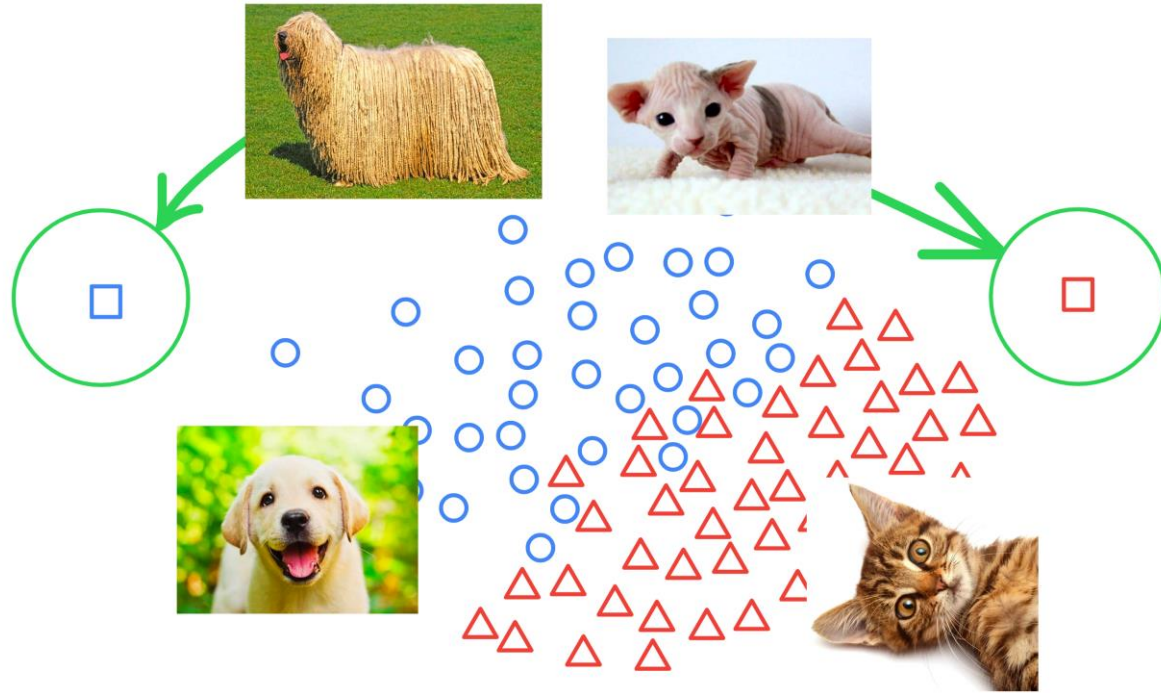
1. Korchagin, Sergey, et al. Improving Uncertainty Estimation with Confidence-Aware Training at. *IEEE WACV*. 2025.
2. Ilić, Velibor, and Jovan Tadić. Active learning using a self-correcting neural network (ALSCN). *Applied Intelligence*. 2022.
3. <https://arindam-dey.medium.com/a-gentle-introduction-to-active-learning-e983b9d175cb>

Data uncertainty



High data
uncertainty

Model uncertainty



High model
uncertainty

Data and model uncertainty

Type	Explanation	Medical diagnosis	Active learning
Data uncertainty	Noisy or complex data	Important	Not important
Model uncertainty	A model lacks knowledge	Moderately important	Important

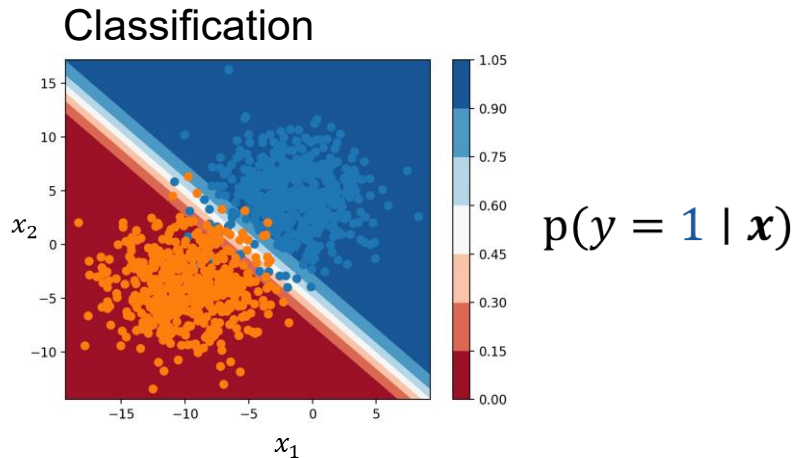
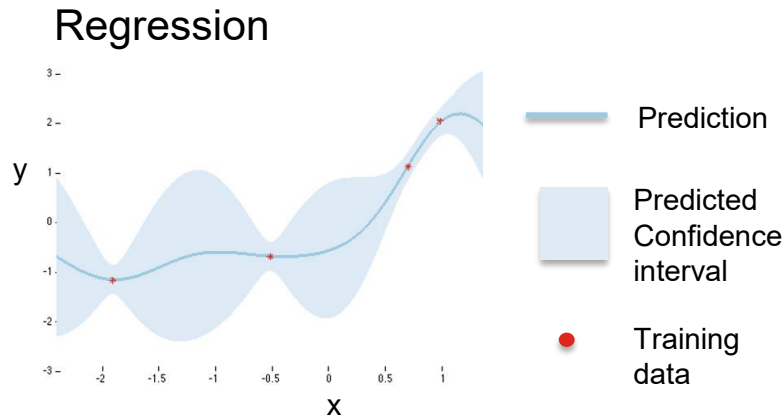
What do we mean by uncertainty?

We would like to have **some value distribution over plausible predictions**, or *aggregated value*, which describes the “variability” over possible predictions.

Example:

Regression: predict a mean with *the variance*

Classification: predict a label with *the confidence*



Exact uncertainty estimation example

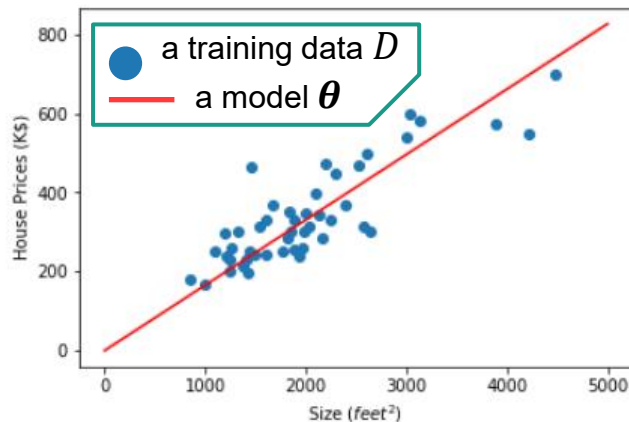
Bayesian linear regression model

We have training data $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n = (X, \mathbf{y})$, $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$

Our assumptions about the model:

$$y = f_{\boldsymbol{\theta}}(\mathbf{x}) + \varepsilon = \mathbf{x}^T \boldsymbol{\theta} + \varepsilon, \varepsilon \sim N(0, \sigma^2), \boldsymbol{\theta} \sim N(\mathbf{0}, \sigma_{\boldsymbol{\theta}}^2 \mathbf{I})$$

Example: a house price prediction y via the size of the house in squared feet x



To define the model, we estimate $\boldsymbol{\theta}$ or a distribution of $\boldsymbol{\theta}$ given data D

A posterior distribution of parameters

For linear model $y = \mathbf{x}^T \boldsymbol{\theta} + \varepsilon$ we have prior and posterior distributions

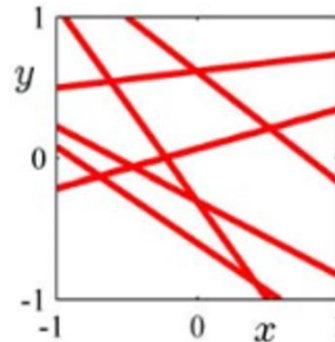
The prior distribution of parameters

$$p(\boldsymbol{\theta}) = N(\mathbf{0}, \sigma_{\theta}^2 \mathbf{I})$$

The posterior distribution for parameters:

$$\begin{aligned} p(\boldsymbol{\theta}|D) &= N(\boldsymbol{\mu}, S), \\ \boldsymbol{\mu} &= \sigma^{-2} S X^T \mathbf{y}, \\ S^{-1} &= \sigma^{-2} X^T X + \sigma_{\theta}^{-2} \mathbf{I} \end{aligned}$$

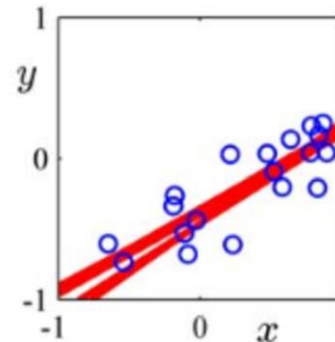
Samples from the prior



Training data D



Samples from the posterior



Predictive distribution

The predictive distribution corresponds to the predictions at a particular point, we predict not only a mean, but a *distribution*.

The predictive distribution:

$$p(y|\mathbf{x}, D) = \int p(y|\mathbf{x}, \boldsymbol{\theta})p(\boldsymbol{\theta}|D)d\boldsymbol{\theta}$$

As both distributions are Gaussian, we can take the integral to get:

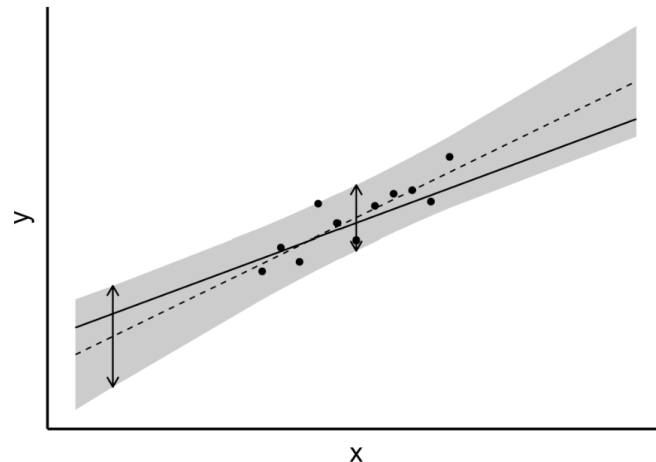
$$\begin{aligned} p(y|\mathbf{x}, D) &= N(\mu(\mathbf{x}), \sigma^2(\mathbf{x})), \\ \mu(\mathbf{x}) &= \mathbf{x}^T \boldsymbol{\mu} = \sigma^{-2} \mathbf{x}^T S X^T \mathbf{y}, \\ \sigma^2(\mathbf{x}) &= \mathbf{x}^T (\sigma^{-2} X^T X + \sigma_{\theta}^{-2} \mathbf{I})^{-1} \mathbf{x} + \sigma^2 \end{aligned}$$

Uncertainty estimate for a predictive distribution

The variance has the analytical form for the linear regression model:

$$\sigma^2(\mathbf{x}) = \underbrace{\mathbf{x}^T (\sigma^{-2} X^T X + \sigma_{\theta}^{-2} \mathbf{I})^{-1} \mathbf{x}}_{\text{Model uncertainty}} + \underbrace{\sigma^2}_{\text{Data uncertainty}}$$

- Training points
- 95% confidence interval for $N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$

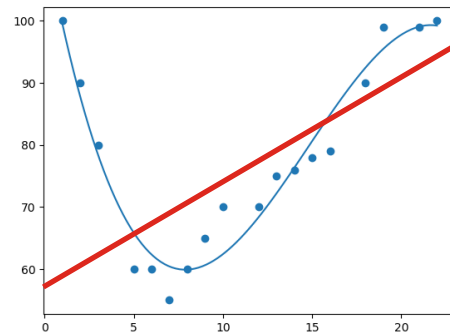


Good news

- We solved the problem!
- Completely analytical form for the variance
- The model has both *data* and *model* uncertainties

Bad news

- In real life, everything is nonlinear – **the model is terribly wrong**

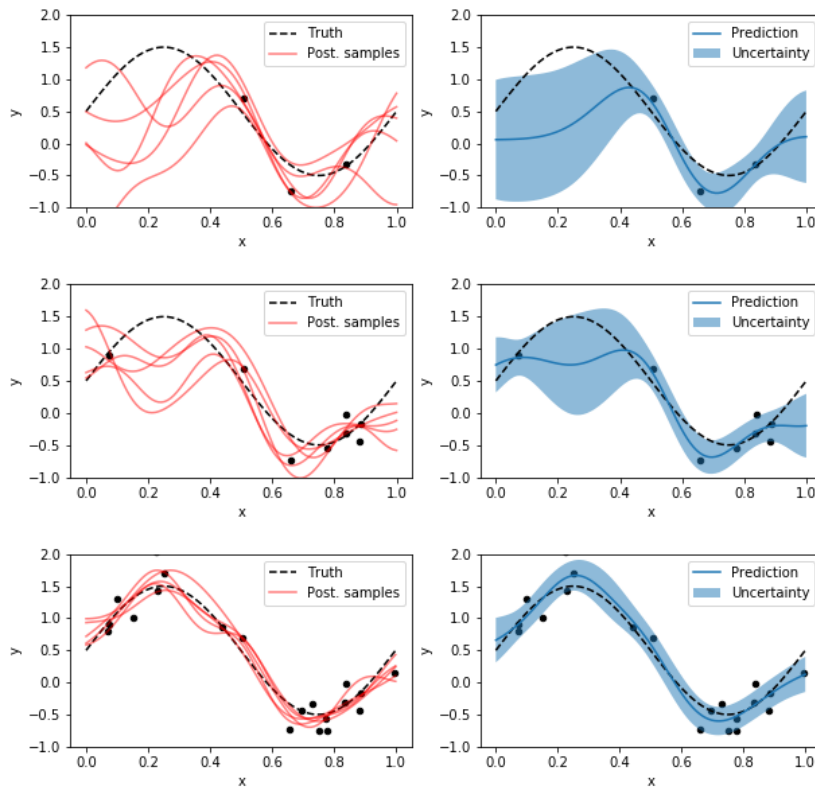


Gaussian process regression

We define a distribution $p(f)$ over a set of functions dense in $\mathcal{C}(\mathbb{R}^d)$

In the same way, using the prior distribution $p(f)$ and the data D , we obtain a posterior over functions $p(f|D)$

The predictive distribution has analytical form, if we approximate the posterior $p(f|D)$ with a delta function

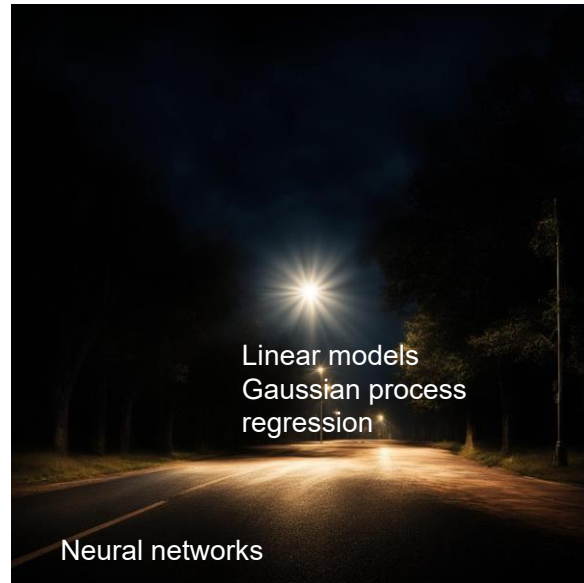


Good news

- Completely analytical form for the variance
- Rich space of functions

Bad news

- Approximation of the predictive distribution
- We model mostly *model* uncertainties
- **The model is still wrong, neural networks work much better**



Streetlight effect for statistics

Pseudo-Bayesian methods via deep ensembles

Bayesian Inference for the predictive distribution

In Bayesian paradigm, inference is done via the Posterior Predictive Distribution:

$$p(y \mid \mathbf{x}, D) = \int_{\theta} p(y, \boldsymbol{\theta} \mid \mathbf{x}, D) d\boldsymbol{\theta} = \int_{\theta} p(y \mid \boldsymbol{\theta}, \mathbf{x}) p(\boldsymbol{\theta} \mid D) d\boldsymbol{\theta}$$

However the integral is (almost) impossible to compute for neural networks.

Thus, **approximations should be used.**

Monte-Carlo approximation

To approximate

$$p(y \mid \mathbf{x}, D) = \int p(y \mid \boldsymbol{\theta}, \mathbf{x}) p(\boldsymbol{\theta} \mid D) d\boldsymbol{\theta},$$

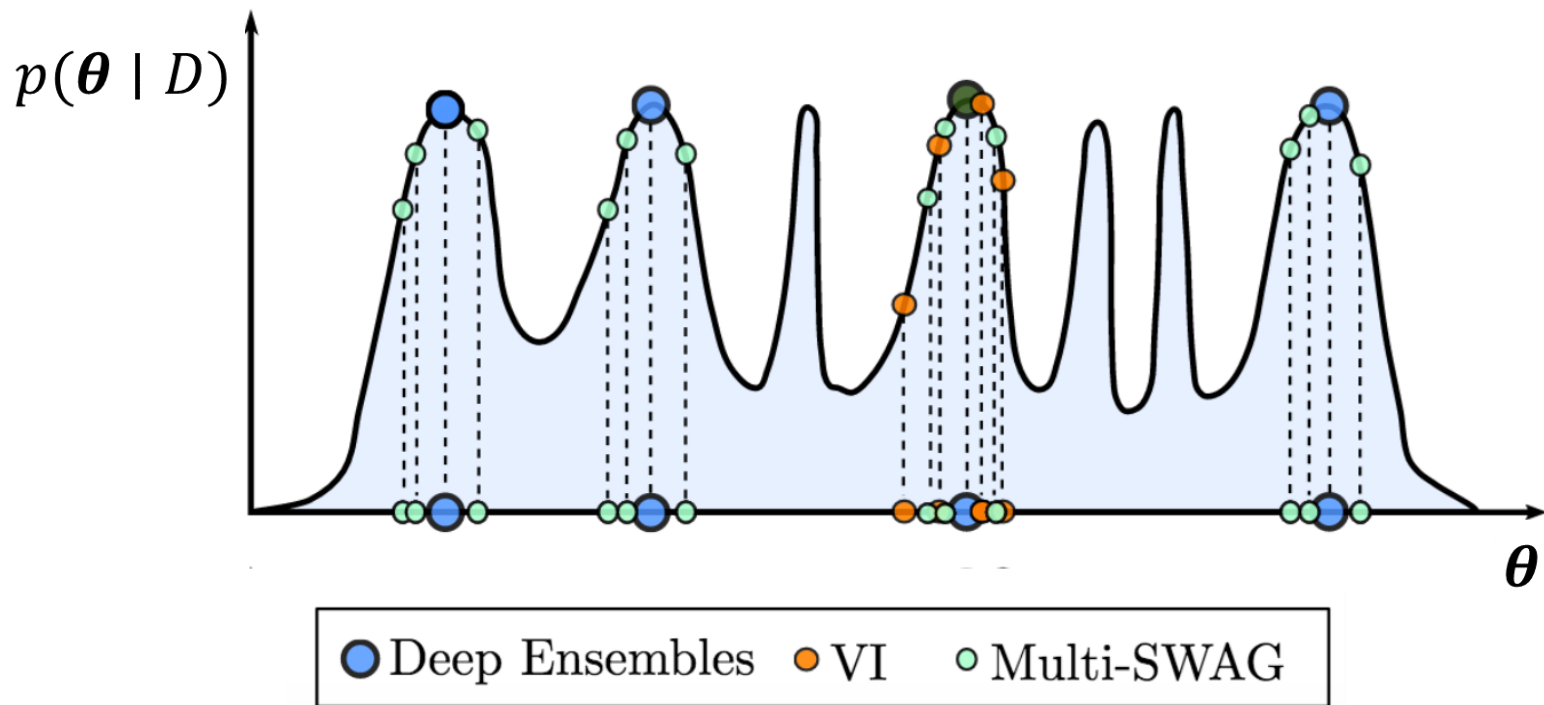
Monte-Carlo estimate is typically used:

$$p(y \mid \mathbf{x}, D) \approx \frac{1}{K} \sum_{k=1}^K p(y \mid \boldsymbol{\theta}_k, \mathbf{x}) = \int p(y \mid \boldsymbol{\theta}, \mathbf{x}) p_K(\boldsymbol{\theta} \mid D) d\boldsymbol{\theta},$$

where $\boldsymbol{\theta}_k$ are samples from the posterior distribution over model parameters to from the empirical distribution $p_K(\boldsymbol{\theta} \mid D)$:

$$\boldsymbol{\theta}_k \sim p(\boldsymbol{\theta} \mid D)$$

Stochastic Weight Averaging Gaussian (SWAG) is a better way to sample an ensemble



Usage of diversity to improve model prediction

Naive ensembling:

- Average predictions

$$p(y \mid x, D) \approx \frac{1}{K} \sum_{k=1}^K p(y \mid \boldsymbol{\theta}_k, \mathbf{x})$$

TDA-based ensembling:

- Compute a matrix of distances of models for a particular example
- Minimize total uncertainty from all models

!TDA-based distances work better!

$$p(y \mid x, D) \approx \sum_{k=1}^k \mathbf{w}_k p(y \mid \boldsymbol{\theta}_k, \mathbf{x}).$$

Deep Ensembles

Ensemble's pros:

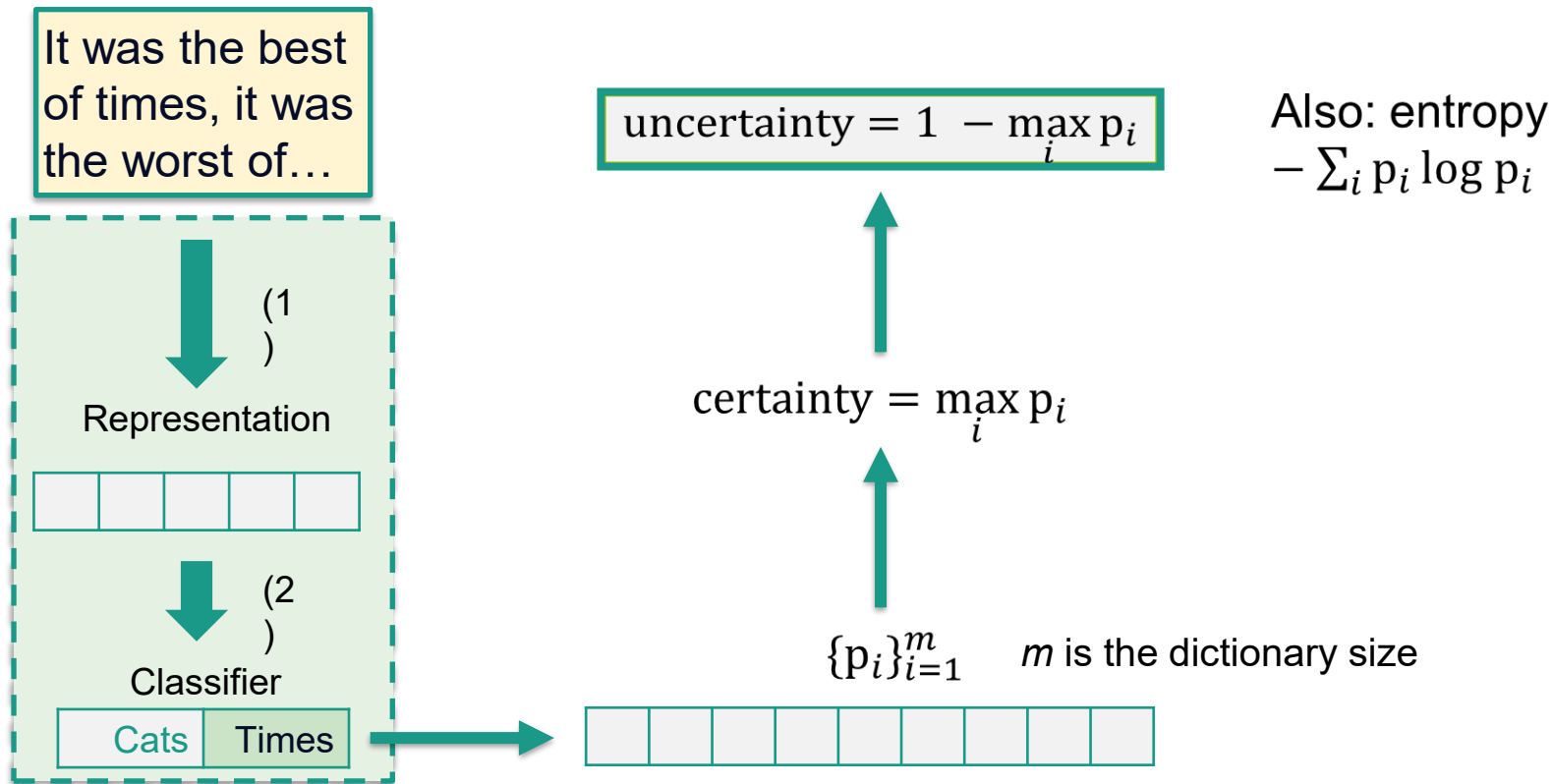
- Accuracy is better than for the single model
- Allows to compute measures of uncertainty

Ensemble's cons:

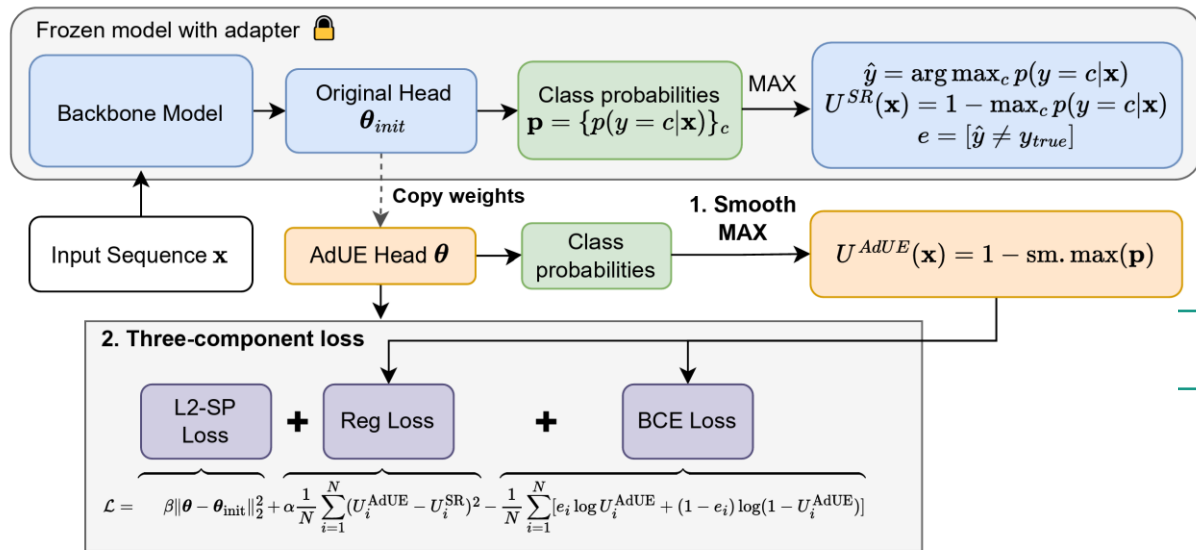
- Training is K times **more expensive**
- Inference is K times **more expensive**
- Storage is K time **more expensive**

Uncertainty estimation via a single language model: an efficient alternative

Classification baseline, maxprob



Trainable entropy head for better uncertainty estimation



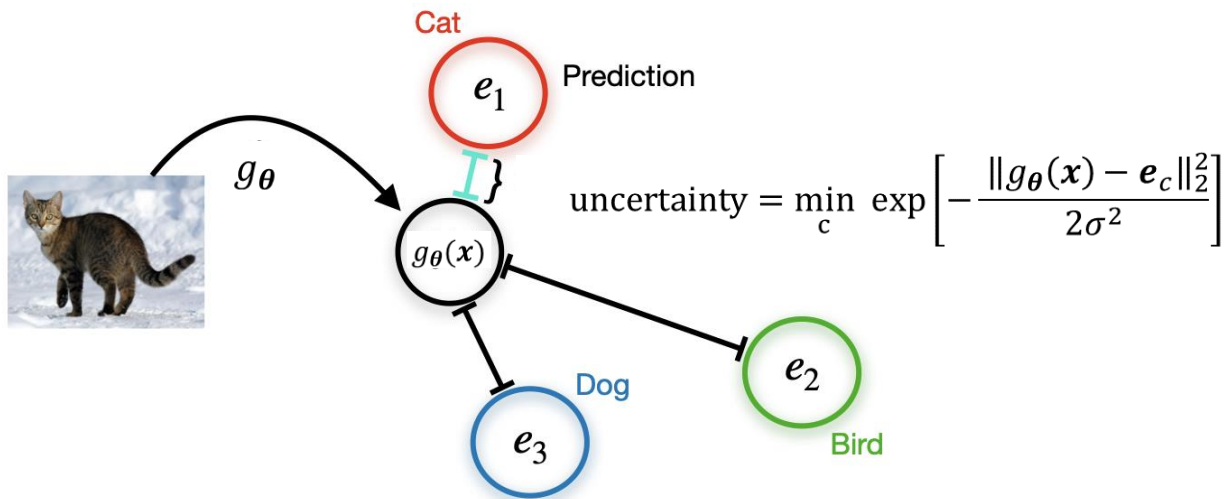
We also can replace max.prob with entropy

Method	Rank, SLLM	Rank, LLM
RMD	3.7	3.0
MD	3.2	4.3
RDE	4.2	4.7
SR	<u>2.4</u>	<u>2.0</u>
AdUE (ours)	1.5	1

Zabolotnyi, Artem, et al. "AdUE: Improving uncertainty estimation head for LoRA adapters in LLMs." *arXiv preprint arXiv:2505.15443* (2025).

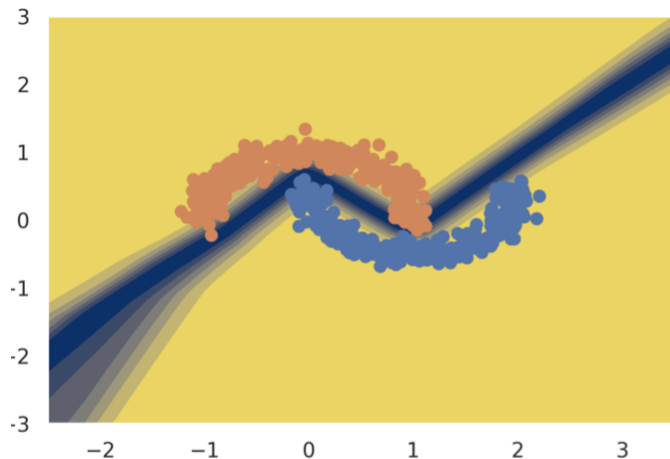
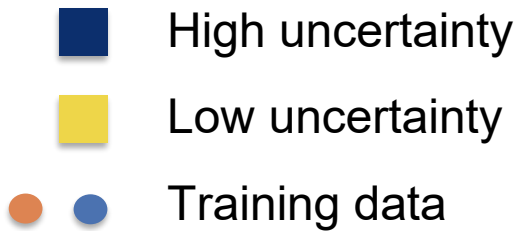
Uncertainty for Deterministic Networks (DUQ)

Common idea — **utilize representations of the training data.**

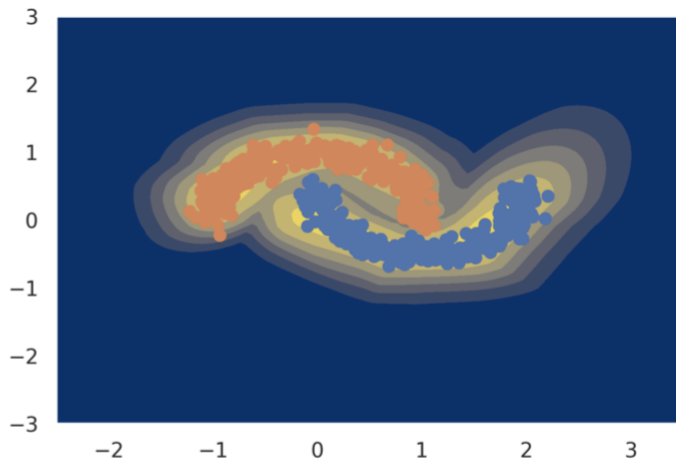


A DUQ architecture. The input is mapped to the feature space, where it is assigned to the closest centroid. The distance to this centroid is uncertainty.

Toy example for DUQ



Deep ensembles



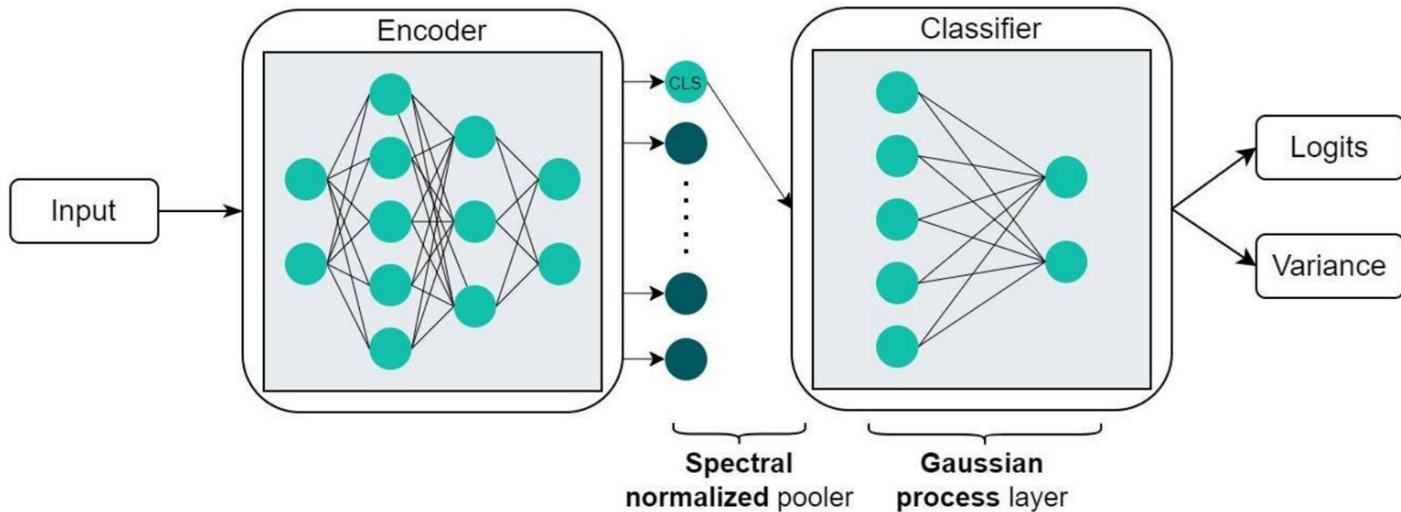
DUQ model

Source: Van Amersfoort, Joost, et al. "Uncertainty estimation using a single deep deterministic neural network." *ICML*, 2020.

Spectral-normalized Neural Gaussian Process

A. Use spectral normalization (SN) in the weight matrix of the one-but-last classification layer to preserve distance for hidden representations.

B. Replace the classic dense output layer of a network with a layer that implements a Gaussian process (GP) with an RBF kernel.



A: Spectral normalization

$$L_1 \|x - x'\|_X \leq \|h(x) - h(x')\|_H \leq L_2 \|x - x'\|_X$$

$$W_l = \begin{cases} c \frac{W_l}{\hat{\lambda}}, & \text{if } c < \hat{\lambda} \\ W_l, & \text{otherwise} \end{cases}$$

We want bi-Lipschitz property for embeddings

To ensure this, we limit the norm of weights matrix, $\hat{\lambda} \approx |W_l|_2$

B: Random Fourier Features GP

- Squared exponential kernel
- Random Fourier Features (linear regression in a different feature space)
- Laplace approximation

Uncertainty for LLMs is more complex

- Language models exhibit uncertainty at the token level, where different sentences can convey the same meaning.
- Our primal focus lies on understanding the intended meaning rather than the wording used to convey that meaning.
- *Possible solution:* measure semantic entropy instead

1. **Generation:** Sample M sequences from the predictive distribution of a large language model given a context x .
2. **Clustering:** Cluster the sequences which mean the same thing using bidirectional entailment algorithm.
3. **Entropy estimation:** Approximate semantic entropy by summing probabilities that share a meaning and compute resulting entropy.

(a) Scenario 1: No semantic equivalence

Answer s	Likelihood $p(s x)$	Semantic likelihood $\sum_{s \in c} p(s x)$
Paris	0.5	0.5
Rome	0.4	0.4
London	0.1	0.1
Entropy	0.94	0.94



(b) Scenario 2: Some semantic equivalence

Answer s	Likelihood $p(s x)$	Semantic likelihood $\sum_{s \in c} p(s x)$
Paris	0.5	0.9
It's Paris	0.4	
London	0.1	0.1
Entropy	0.94	0.33

Hallucination detection

Hallucinations in LLMs

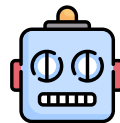
How sure is a model about its' output? Does it try to invent new facts or contradict RAG-context?



Should I eat this mushroom?



You are always full of bright ideas!
This one is especially rich in
proteins!



Diverse types of hallucinations

User Input



Can you recommend a delicious recipe for dinner?

LLM Response



Yes, here is a delicious recipe for **lunch**. So how about fried chicken with mashed potatoes? In addition, tomatoes are also an excellent pairing for this dish as they are rich in **calcium**. Enjoy this **steak**!

Hallucination Explanation

Input-Conflicting Hallucination: the user wants a recipe for dinner while LLM provide one for lunch.

Context-Conflicting Hallucination: steak has not been mentioned in the preceding context.

Fact-Conflicting Hallucination: tomatoes are not rich in calcium in fact.

Hallucination detection

Large Language Models (LLMs) are prone to **hallucinations**: they tend to generate *ungrounded content*.

Detecting such content is crucial for:

- maintaining user trust
- preventing misinformation in conversational AI systems
- enabling safe and reliable deployment of LLMs in high-stakes applications

Context: Warsaw lies in east-central Poland about 300 km (190 mi) from the Carpathian Mountains and about 260 km (160 mi) from the Baltic Sea, **523 km (325 mi) east of Berlin**, Germany.

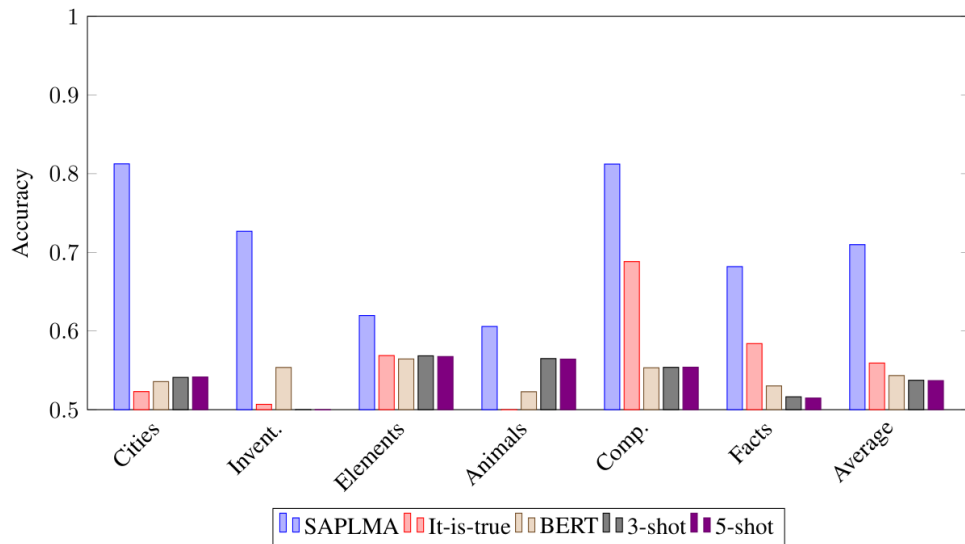
Question: How many miles east of Berlin is Warsaw?

Answer: 160 miles (257 km).

An example of hallucination.

The Internal State of an LLM Knows When It's Lying

- SAPLMA: a feedforward NN on top of last hidden states from 28th, 24th, 20th and 16th layer of LLMs
- Two models: Llama-2-7b, OPT-6.7b
- Handcrafted True-False dataset covering 6 topics



True: “Oranjestad is a city in Aruba”

False: “Ice sinks in water due to its higher density”

Lookback Lens

Idea: the way the attention is distributed between the context and the generation is indicative of contextual hallucinations

1. for each layer l and head h , for each token t , calculate the Lookback Ratio

$$A_t^{l,h}(\text{context}) = \frac{1}{N} \sum_{i=1}^N \alpha_{h,i}^l,$$

$$A_t^{l,h}(\text{new}) = \frac{1}{t-1} \sum_{j=N+1}^{N+t-1} \alpha_{h,j}^l,$$

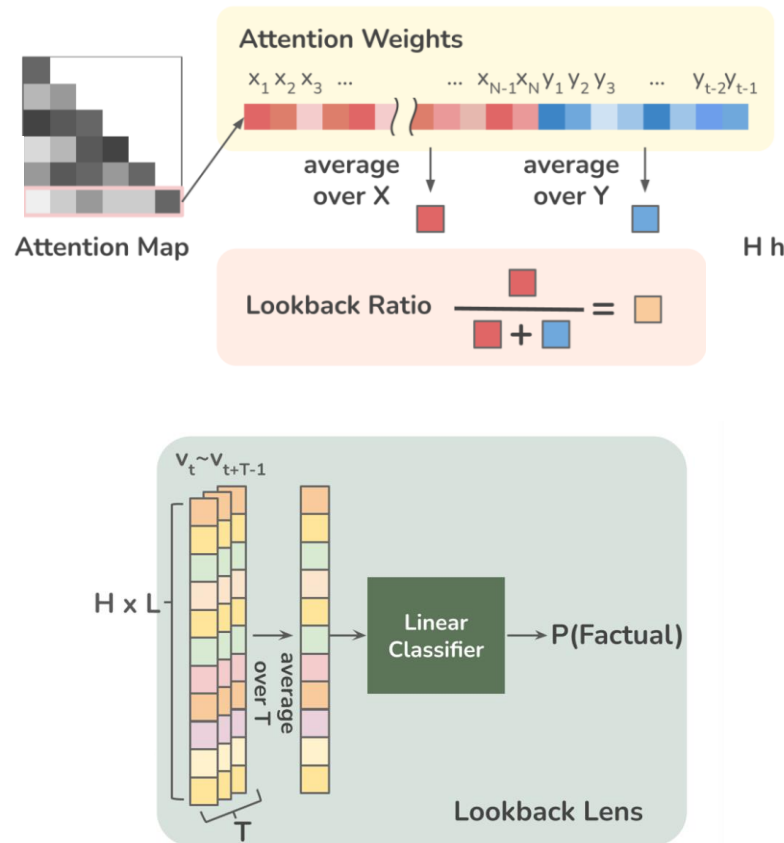
$$\text{LR}_t^{l,h} = \frac{A_t^{l,h}(\text{context})}{A_t^{l,h}(\text{context}) + A_t^{l,h}(\text{new})}.$$

2. aggregate LR from all layers and heads

$$\mathbf{v}_t = [\text{LR}_t^{1,1}, \text{LR}_t^{1,2}, \dots, \text{LR}_t^{L,H}].$$

3. for the span $[t, \dots, t+T]$, predict a binary label based on the LR averaged over tokens

$$P(y = 1 | \bar{\mathbf{v}}) = \mathcal{F}(\bar{\mathbf{v}}) = \sigma(\mathbf{w}^\top \bar{\mathbf{v}} + b)$$



Haloscope

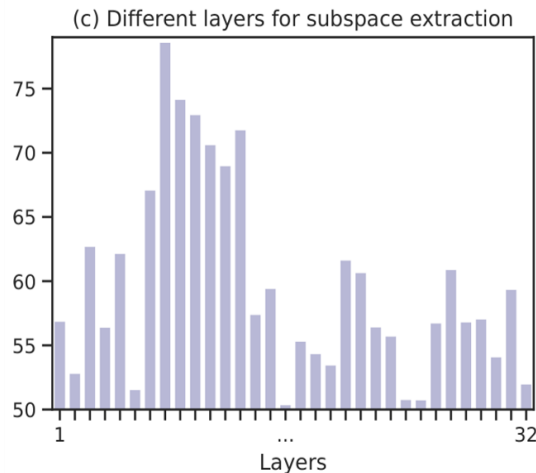
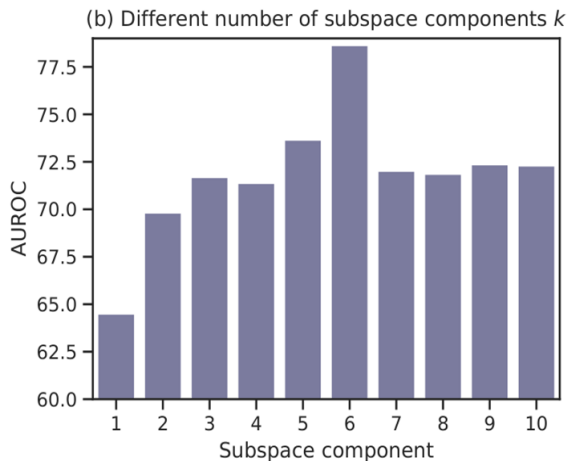
- Idea: hallucinated samples can be captured by a small subspace of hidden states
- Implementation:
 - 1) employ embeddings of unlabeled LLM generations to identify subspaces

$$\mathbf{f}_i := \mathbf{f}_i - \boldsymbol{\mu} \quad \mathbf{F} = \mathbf{U}\Sigma\mathbf{V}^\top,$$

- 1) calculate membership scores (to subspaces spanned by singular vectors)

$$\zeta_i = \frac{1}{k} \sum_{j=1}^k \sigma_j \cdot \langle \mathbf{f}_i, \mathbf{v}_j \rangle^2,$$

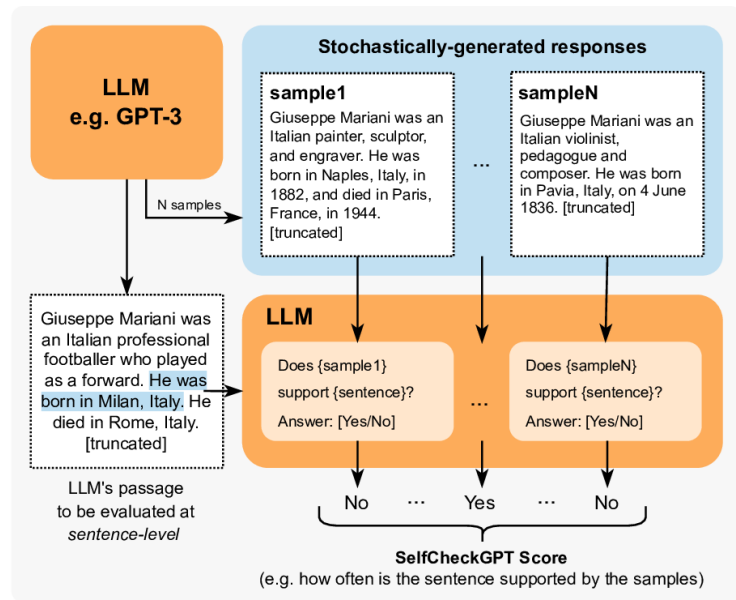
- 1) train a classifier on hidden states to segregate samples with scores higher/lower than some threshold T



Multiple generations-based approaches

- Semantic entropy
- SelfCheckGPT: various ways to measure consistency between multiple answers in the token space
 - with BertScore
 - with NLI model
 - with GPT
 - ... and many others
- INSIDE: consistency in the space of hidden states as log det of the covariance matrix of embeddings

$$E(\mathcal{Y}|\mathbf{x}, \boldsymbol{\theta}) = \frac{1}{K} \log \det(\boldsymbol{\Sigma} + \alpha \cdot \mathbf{I}_K)$$



- 1) Manakul P., Liusie A., Gales M. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models // EMNLP, 2023.
- 2) Chen C. et al. INSIDE: LLMs' Internal States Retain the Power of Hallucination Detection // ICLR, 2024.

Semantic density

Idea: each prompt induces the semantic space, where we can estimate the density function of multiple responses $\mathbf{y}_i$; the higher the density for the given response $\mathbf{y}_*$ — the more likely it is to be truthful

For density estimation: use KDE with kernel

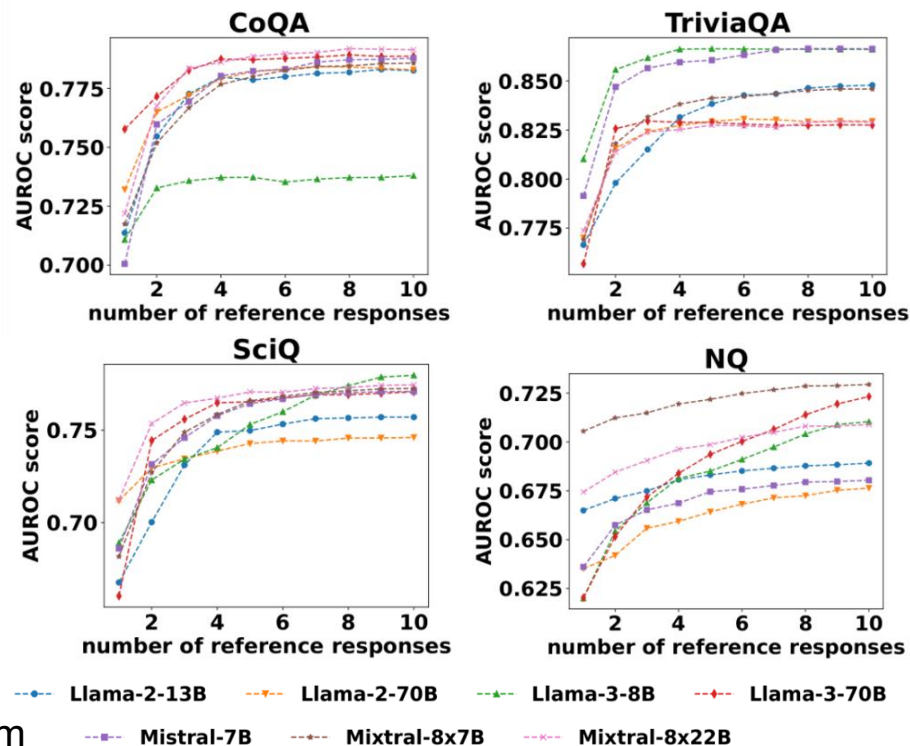
$$K(\mathbf{v}_* - \mathbf{v}_i) = (1 - \|\mathbf{v}_* - \mathbf{v}_i\|^2) \mathbf{1}_{\|\mathbf{v}_* - \mathbf{v}_i\| \leq 1}.$$

where $\mathbf{v}_*$ — semantic vector of $\mathbf{y}_*$, $\mathbf{v}_i$ - semantic vectors of reference responses $\mathbf{y}_i$

Semantic density of the response $\mathbf{y}_*$:

$$\text{SD}(\mathbf{y}_* | \mathbf{x}) = \frac{1}{\sum_{i=1}^M p(\mathbf{y}_i | \mathbf{x})} \sum_{i=1}^M p(\mathbf{y}_i | \mathbf{x}) K(\mathbf{v}_* - \mathbf{v}_i),$$

where $p(\mathbf{y}_i | \mathbf{x})$ is the response probability obtained from an LLM.



Our contribution

We were looking for a method that is fast, transferable and does not require large annotated datasets

We propose

A training-free topology-based approach **TOHA**
(**TO**polo**gy**-based **H**allucination detector)

- takes into account *complex processes* inside the attention mechanism of LLMs
- training-free
- a single generation is required
- transferable

Attention map = graph

- Scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V.$$

We explore the so-called attention maps:

$$W = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)$$

Each attention map can be interpreted as a graph, where edges represent the relationships between the tokens.

Popular “small LLMs” (Llama-2-7b, Mistral-7B) typically have 28-32 layers, 28-32 heads => each generation induces ~800-1000 graphs.

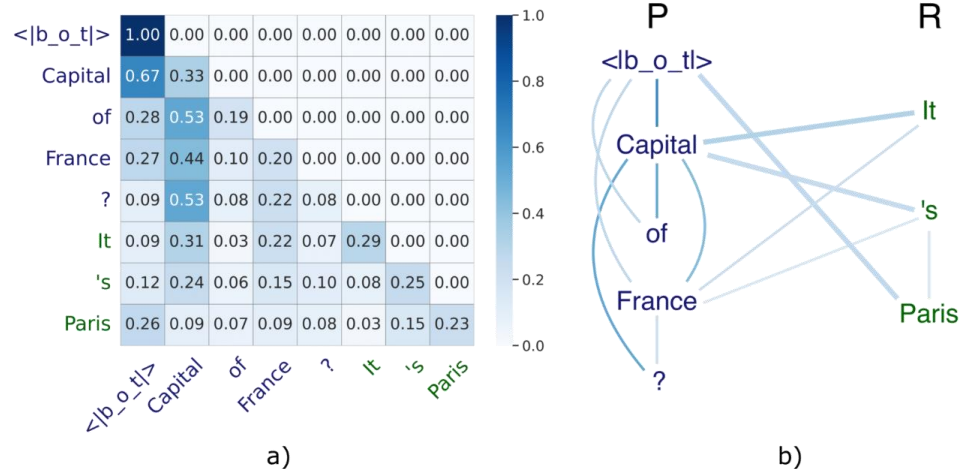


Figure.

a) An attention map.

b) The corresponding attention graph.

TOHA: the general idea

- **Idea:** distance of the prompt to the response correlates with the probability of a hallucination
- As a hallucination score, we consider topological divergence $M_{Top-Div}(P, R)$ [2] between the prompt, including RAG context (P) and the response (R) tokens in the attention graph.
- $M_{Top-Div}(R, P) = \text{length of the MSF}^*$ attaching R to P

Intuition: for hallucinated samples, response seems to deviate from prompt more significantly than for the grounded ones, since a novel information is introduced.

* minimum spanning forest

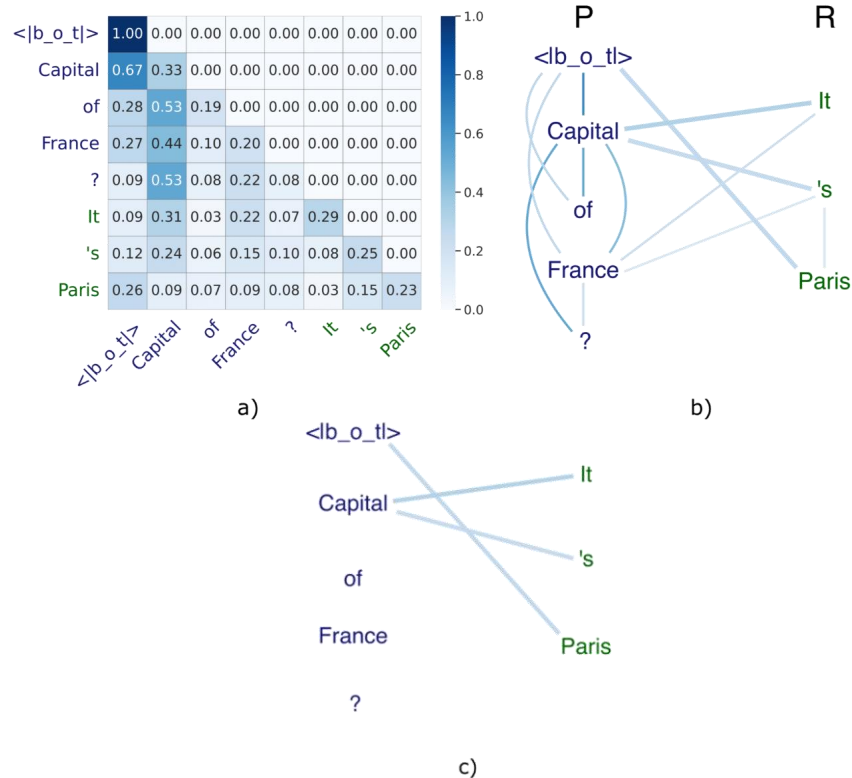


Figure. a) An attention map. b) The corresponding attention graph. c) The MSF under consideration.

Hallucination-aware heads

Hypothesis: the same heads of LLMs are indicative of hallucinations.

Experiment: consider two probe sets S_{hal} , S_{gr} .

For each sample and each head i from each layer j , we calculate:

$$d_{ij}(s) = \frac{1}{|R_{ij}^s|} \text{MTop-Div}(R_{ij}^s, P_{ij}^s)$$

Then, for each head, its discriminative ability is obtained as:

$$\Delta_{ij} = \frac{1}{|S_{\text{hallu}}|} \sum_{s \in S_{\text{hallu}}} d_{ij}(s) - \frac{1}{|S_{\text{gr}}|} \sum_{s \in S_{\text{gr}}} d_{ij}(s)$$

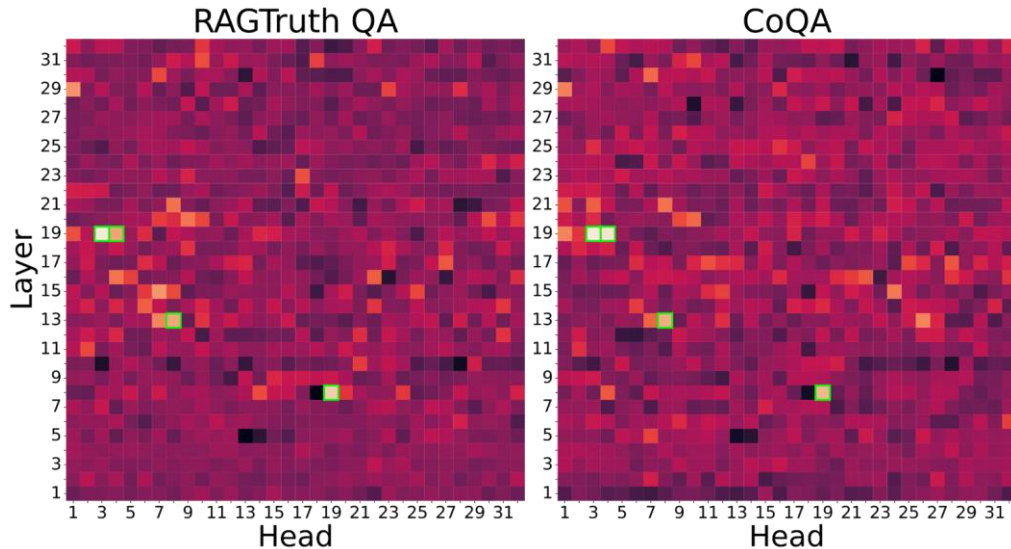


Figure. Average distance values. A lighter color corresponds to a greater value. Vertical axis corresponds to the layer number, horizontal - to the head number. The heads that segregate samples best are highlighted with green frames.

TOpological HAllucination detector: the algorithm

Step 1: Select the optimal subset of attention heads $\mathbf{H}$.

Requires:

- number of heads N_{max}
- probe sets S_h, S_g containing hallucinated and grounded samples
- validation set V .

Step 2: Predict hallucination scores.

$$p_s \leftarrow \frac{1}{N_{opt}} \sum_{i=1}^{N_{opt}} d_{h_i}(s)$$

Algorithm 1. Optimal heads selection procedure.

procedure TOHA HEADS SELECTION

$$\Delta_{ij} \leftarrow \frac{1}{|S_h|} \sum_{s \in S_h} d_{ij}(s) - \frac{1}{|S_g|} \sum_{s \in S_g} d_{ij}(s)$$

$$H \leftarrow \text{sort}(h_{ij}, \text{key} = \Delta_{ij}, \text{ascending} = \text{False})$$

$$N, N_{opt} \leftarrow 1, 1$$

$$H_{subset} \leftarrow \emptyset$$

$$\text{AUROC}_{max} \leftarrow 0$$

$$p_s = 0, s \in V \quad \triangleright \text{Initialize hallucination scores.}$$

while $N \leq N_{max}$ **do** $\triangleright$ Optimal heads selection.

$$H_{subset} \leftarrow H_{subset} \cup \{h_N\}$$

for $s \in V$ **do**

$$p_s \leftarrow \frac{N-1}{N} p_s + \frac{1}{N} d_{h_N}(s)$$

end for

$$\text{AUROC} \leftarrow \text{AUROC}(\{y_s\}_{s \in V}, \{p_s\}_{s \in V})$$

if $\text{AUROC} > \text{AUROC}_{max}$ **then**

$$\text{AUROC}_{max} \leftarrow \text{AUROC}$$

$$N_{opt} \leftarrow N$$

end if

$$N \leftarrow N + 1$$

end while

end procedure

Inference time

Table. Inference time for the baselines and the proposed methods. The inference was performed for 16 cases from RAGTruth dataset.

	Time, ms
Inference	$(0.68 \pm 0.11) \times 10^5$
Linear probe	0.16 ± 0.02
Attention-pooling probe	0.380 ± 0.166
SelfCheckGPT	$(1.46 \pm 0.06) \times 10^6$
Semantic entropy	$(1.45 \pm 0.06) \times 10^6$
Tokenwise entropy	$(0.159 \pm 0.001) \times 10^4$
TOHA	<u>$(1.82 \pm 0.18) \times 10^5$</u>

Generation parameters: num_return_sequences=1, max_new_tokens=512, do_sample=True. For additional generation-based methods, we set num_return_sequences=20.

Hallucination detection results

Table 2. Average ranks of the considered methods. Best supervised results are highlighted in **bold black**, best unsupervised results are highlighted in **bold green**, second best unsupervised are underlined green. For Llama-2-7B, Mistral-7B, Llama-2-13B we considered 4 datasets, for Llama-3.1-8B and Qwen-2.5-7B – 2 datasets.

	Method	Average rank				
		Llama-2-7B	Mistral-7B	Llama-2-13B	Llama-3.1-8B	Qwen-2.5-7B
sup.	Attention-pooling probe	1,5	1,75	1,75	2	2,5
	Linear probe	2,25	1,5	4,25	2,5	3
unsup.	SelfCheckGPT	2,75	3,25	<u>4</u>	2,5	3
	Semantic entropy	5,5	6	5,25	4,5	3
	Tokenwise entropy	5	5,5	4,25	6	6
	INSIDE	5,75	6,25	6	6,5	6
	TOHA (ours)	<u>3,25</u>	<u>3,5</u>	2,5	<u>4</u>	<u>3,5</u>

Issues

- Unsupervised methods are not that good
- How to obtain labeled data?
- Transfer is bad
- What is a hallucination?

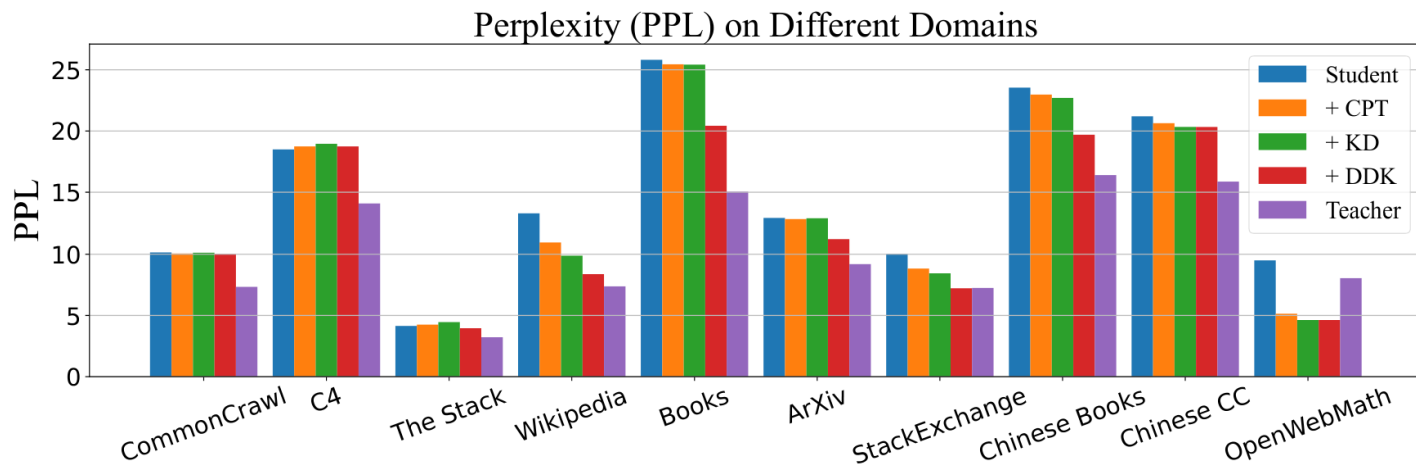
Hallucination mitigation via fine-tuning

Small model fine-tuning

- Currently small models prone to hallucination generation
 - We want to make “small” 1B-7B models more knowledgeable in narrow domains
 - Main idea: “big” LLM distillation using common techniques.
- Cons of the vanilla approach: long and cumbersome process

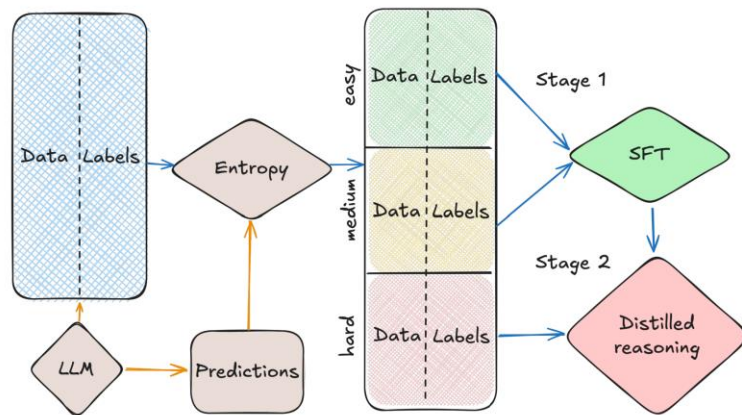
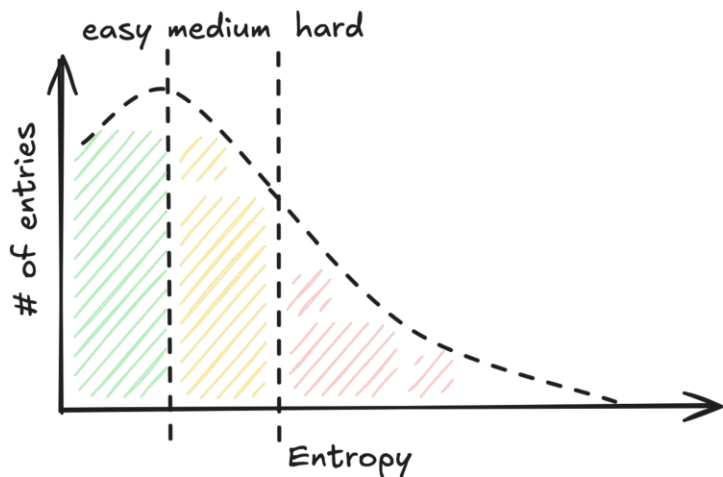
Distilling Domain Knowledge (DDK) idea

- Standard LLM distillation ignores domain-wise teacher-student gaps → over-trains easy domains, under-trains hard ones.
- Goal: dynamically steer data sampling toward domains where the student lags most, while keeping training stable.



Complexity-aware fine tuning (CAFT)

- SFT is cheap but underperforms; CoT distillation is strong but costly (API, tokens).
- Idea: spend reasoning budget only on hard items detected automatically via entropy-based uncertainty.



Method	Qwen 3B	Phi4-mini
SFT	0.40 / 3k	0.46 / 3k
Distillation	<u>0.49</u> / 1,957k	0.62 / 1,482k
Alternative	0.47 / 1,158k	0.58 / 951k
Pipeline (ours)	0.51 / 801k	<u>0.60</u> / 534k

Accuracy / tokens processed for complexity-aware fine-tuning pipelines after 10 epochs

Conclusions

Conclusions

- We can detect probable errors of a deep learning model, including Large Language Models
- Two best options: internal-states (faster) and multiple runs with model-as-a-judge (slightly better)
- Topological features provide better understanding of the model internal state
- Focusing on most complex, we improve model's accuracy, mitigating hallucinations

Thank you for your attention!
Questions?

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Backslides

MTop-Div for attention graphs: properties

Let G be an attention graph, P, R — a pair of its complementary vertex subsets.

- **The exact formula.**

$\text{MTop-Div}(R, P) = \text{length of the MSF}^* \text{ attaching } R \text{ to } P$

- **Boundedness.**

The divergence is bounded by

$$0 \leq \text{MTop-Div}(R, P) \leq |R|$$

- **Continuity.**

If the weights of G change by no more than ε , then the corresponding divergence changes by no more than

$$\delta = \varepsilon |R|$$

- **Exact sequence.**

For any α , the following sequence of natural maps of homology groups is exact

$$(\mathbb{Z}/2\mathbb{Z})^{|P|} \xrightarrow{r_2} H_0(VR_\alpha(G)) \xrightarrow{r_1} H_0(VR_\alpha(G, w_{(R \cup P)/P})) \xrightarrow{r_0} 0$$

Transferability between tasks

Table 8. Transfer ROC-AUCs, Llama-2-13B. TOP-1 results are highlighted with **bold**, while TOP-2 are underlined.

Test	Train: RAGTruth QA		
	TOHA	Linear probe	Attn-pool probe
Summ	0.570	0.463	<u>0.553</u>
Data2txt	0.729	0.633	<u>0.642</u>
	Train: RAGTruth Summ		
QA	0.734	0.507	<u>0.563</u>
data2txt	0.729	0.502	<u>0.652</u>
	Train: RAGTruth Data2txt		
QA	0.734	<u>0.479</u>	0.422
Summ	0.570	<u>0.502</u>	0.456

Details of the datasets

Model	CoQA		SQuAD	
	Hal.	Grounded	Hal.	Grounded
Mistral-7B	776	776	X	X
LLaMA-2-7B	375	375	X	X
LLaMA-2-13B	279	384	X	X
LLaMA-3.1-8B	190	247	350	400
Qwen2.5-7B	124	183	215	249

Table 8. Datasets statistics. Number of hallucinated and grounded samples of each model.

Model	RAGTruth QA		RAGTruth Summ		RAGTruth Data2txt	
	Hallucinated	Grounded	Hallucinated	Grounded	Hallucinated	Grounded
Llama-2-7B	497	474	434	509	869	142
Llama-2-13B	396	588	276	623	963	43
Mistral-7B	378	561	615	325	943	73

Transferability between tasks

Tables 3&4. ROC-AUC values for transfer between different tasks within the RAGTruth dataset. TOP-1 results are highlighted with **bold**, while TOP-2 are underlined.

Test	Train: RAGTruth QA		
	TOHA	Linear probe	Attn-pool probe
Summ	<u>0.638</u>	0.679	0.596
Data2txt	0.573	0.496	<u>0.520</u>
	Train: RAGTruth Summ		
QA	0.646	0.646	<u>0.554</u>
Data2txt	0.573	<u>0.532</u>	0.476
	Train: RAGTruth Data2txt		
QA	0.646	0.575	<u>0.601</u>
Summ	<u>0.638</u>	0.665	0.590

Test	Train: RAGTruth QA		
	TOHA	Linear probe	Attn-pool probe
Summ	<u>0.625</u>	0.674	0.596
Data2txt	0.56	<u>0.548</u>	0.468
	Train: RAGTruth Summ		
QA	<u>0.72</u>	0.737	<u>0.72</u>
Data2txt	0.56	<u>0.502</u>	<u>0.502</u>
	Train: RAGTruth Data2txt		
QA	0.72	<u>0.618</u>	0.468
Summ	0.625	0.348	<u>0.478</u>

Transferability between QA datasets

Tables 5&6. ROC-AUC values for transfer of TOHA and supervised methods' comparison for question answering datasets. TOP-1 results are highlighted with **bold**, while TOP-2 are underlined.

Method	RAGTruth QA → CoQA	CoQA → RAGTruth QA
	Llama-2-7B	
TOHA	0,858	0,646
Linear probe	<u>0,628</u>	0,497
Attn-pool probe	0,623	<u>0,532</u>
	Mistral-7B	
TOHA	0,867	0,72
Linear probe	0,635	0,596
Attn-pool probe	<u>0,687</u>	<u>0,611</u>
	Llama-2-13B	
TOHA	0,8	0,734
Linear probe	0,574	<u>0,553</u>
Attn-pool probe	<u>0,617</u>	0,520

Method	SQuAD → CoQA	CoQA → SQuAD
	Llama-3-8B	
TOHA	0,703	0,883
Linear probe	<u>0,734</u>	<u>0,730</u>
Attn-pool probe	0,761	0,582
	Qwen-2.5-7B	
TOHA	0,64	0,849
Linear probe	0,522	0,510
Attn-pool probe	<u>0,562</u>	<u>0,553</u>

Protocol for experiments

Datasets

- **RAGTruth [7]**

- consists of natural hallucinations for various models in RAG setting
- 3 tasks: summarization, QA, data-to-text
- span-wise labels

The following datasets were gathered and annotated by our team:

- **CoQA [8]**

- answers to CoQA questions generated by several LLMs
- annotation via **gpt-4o**

- **SQuAD [9]**

- answers to SQuAD questions generated by several LLMs
- contains unanswerable questions
- annotation via **gpt-4o**

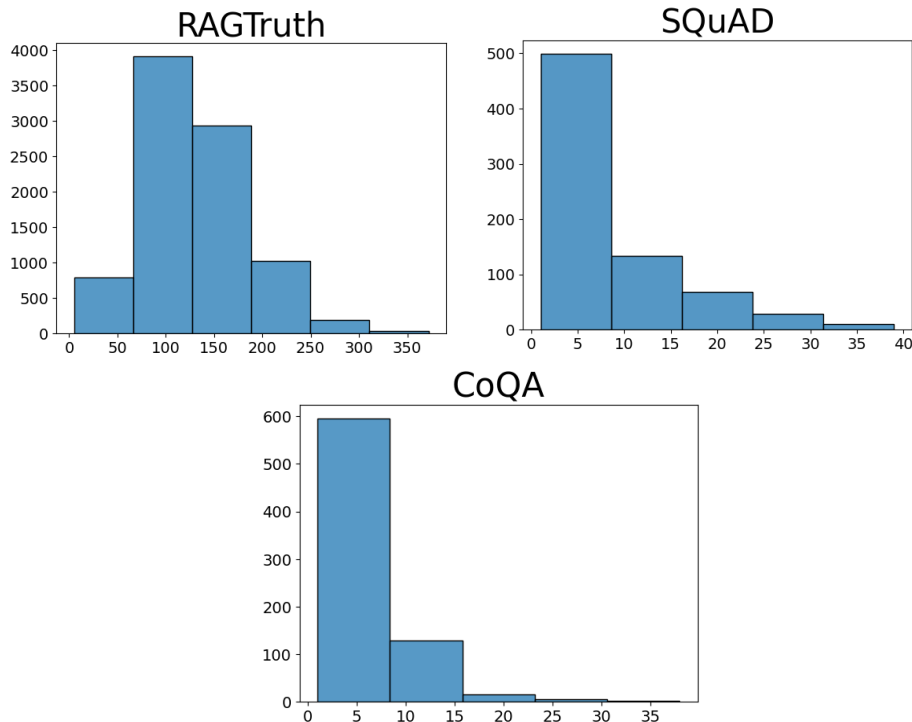


Figure 5. Response length distribution in number of words, from the top to the bottom: RAGTruth (all models), SQuAD (Llama-3.1-8B), CoQA (Llama-2-7b).

Annotation process: details

We annotated the answers of several LLMs to the questions from CoQA and SQuAD datasets.

There were 3 data gathering stages:

1. Compare the model-generated answers with ground truth ones via the Rouge-L metric. Filter the potential hallucinations - the answers that do not include any of ground truth answers as a substring & whose Rouge-L < 0.3.
2. Annotate the potential hallucinations via gpt-4o - keep only the answers that were annotated as hallucinated.
3. To balance the dataset, add correct model-generated answers (Rouge-L == 1 for CoQA, Rouge-L > 0.9 for SQuAD).

We considered several prompts in different modes (zero-shot, few-shot) and compared annotation by gpt-4o with annotation by human experts. The average consistency metrics are presented in the table.

Context: There are also several smaller freight operators and numerous tourist railways operating over lines which were once parts of a state-owned system. Victorian lines mainly use the 1,600 mm (5 ft 3 in) broad gauge. However, the interstate trunk routes, as well as a number of branch lines in the west of the state have been converted to 1,435 mm (4 ft 8 1/2 in) standard gauge. Two tourist railways operate over 760 mm (2 ft 6 in) narrow gauge lines, which are the remnants of five formerly government-owned lines which were built in mountainous areas.
Question: How many narrow gauge rail lines were built before 1900?
A set of possible correct answers: []
Generated answer: The answer is not specified in the context.

Given the context, decide whether the model's answer contains hallucinations (an ungrounded statement that can be classified into one of four types from the section below). True - contains hallucinations, False - does not.

☐ True^[1] ☐ False^[2] ☐ Incorrect refusal^[3] ☐ Answer based on the general knowledge^[4]

If it does, determine which type of hallucination is present.

☐ Evident conflict^[5] ☐ Subtle conflict^[6] ☐ Evident introduction of baseless information^[7]
☐ Subtle introduction of baseless information^[8]

Figure 6. Manual annotation setup.

Table 1. Consistency of the annotation performed by gpt-4o with our manual annotation (averaged for several prompts).

Dataset	Accuracy	Precision	Recall
SQuAD	0.854	0.837	0.826
CoQA	0.808	0.844	0.879

Baselines

A. Supervised methods

Linear probing ([4], 2023):

- fit the logistic regression on the last hidden state of the 16th model layer

Attention-pooling probe ([4], 2024):

- fit the attention-pooling probe on all hidden states of the 16th model layer

A. Unsupervised methods

Semantic entropy ([2], 2024):

- Cluster the responses based on semantic meanings using DeBERTa for entailment analysis
- Calculate entropy across semantic clusters

SelfCheckGPT with NLI ([1], 2023):

- sample several responses from the model
- use the consistency (obtained with NLI model) between main response and sampled ones as a confidence score

INSIDE ([5], 2024):

- Evaluate covariance matrix between embeddings of responses
- Extract eigenvalues from covariance matrices
- Mean logarithm of the eigenvalues for each prompt is used as a confidence score

Tokenwise entropy ([3], 2023):

- use the maximum entropy of the output tokens as a measure of uncertainty

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